

Fundamentals of Machine learning for and with engineering applications

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Course objectives

Understanding of

- data sources and consequent data properties.
- data analysis/machine learning approaches outcomes.
- sensitivity analysis
- predictive modeling
- multivariate data analysis
- machine learning techniques application
- ensemble methods
- Bayes approach
- visualization and reporting

Metalearning

- Pedagogic empiricism teaches us that good learning happens in social contexts where the students are actively participating in the classroom dialogue
- In general, the number of questions, and the dialogue in general, drops significantly when lectures are recorded
- Streaming tends to lead to empty classrooms and higher failure rate.

Teaching method

!Active Learning!

- There will be a combination of lectures and tutorials.
- Tutorial and hands-on will be presented during the course.
- Study groups are strongly encouraged.
- In class discussions are encouraged at any stage.
- Flip classroom approach: problem first (when possible).
- Feedback expected from you!
- Note: each teacher, even if coordinated, will have a different approach/expectations.

Python

The tutorials and hands-on will be mostly based on python.

- An introduction to python will be provided during the course. Yet, this is not a Python programming course!
- You are **strogly advised** to construct your own Git repository (and to keep it in order).
- Group projects will have different roles.

A personal considerations on generative AI

With great power comes great responsibility (Spider-Man)

LLMs can and shall be used. As their development is surging in the last years, their help in writing code and reports is undeniable. Students shall learn how to master these tools. Yet, while their usage is encouraged, the risk of excessively rely on them **DO** hinder learning.

If you want to be a student (i.e. whom is learning), you are encouraged to take the responsibility in delivering original output.

Why are we here? Why this course?

- Decisions are the hinge. What influences your decisions has value.

- Better results come from better decisions.

STATISTICAL argument

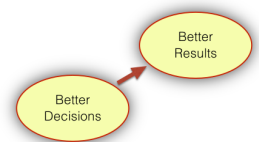
- Machine learning can be a useful **ONLY** when it helps making better decisions.

Decisions, decisions, decisions...

Life lesson?!

Life is a sum of all your choices (Albert Camus)

The only way you can purposefully influence your life, your family, your organization, your country or your world is through **the decisions you make**.



Uncertainty is fed to decisions



- In this course we will learn to use data and models (ML) to make better decisions.
- Each application will be a mere exercise of the concepts we will introduce here.

Do not try this at home!

- Please Do not use a technical approach for love matters.**

Uncertainty

It is present in every stage of life (at each decision).

You would thus need to:

- 1 Rationalize uncertainty (qualify)
- 2 Quantify uncertainty
- 3 Make decisions under uncertainty
- 4 Operate under uncertainty

Uncertainty or Probability?

Our aim here is to provide good guidance on how to link data, models and output to value creation.

- First we need to understand uncertainty and probability, and the difference between the two.

- Spoiler: Probability is the language of uncertainty!

- We will analyse, quantify and structure models as a function of uncertainty.
- Clarity of language in probability is one of the hallmarks of decision analysis and of modeling.

Data properties

- All starts from data: what are data-properties?
- Are there such things as good data and bad data?

Life lesson (or exam question, same thing ;))

- Data **DO NOT** always have value.

- TRASH in TRASH out

Spatial and Temporal Data

Statistics is collecting, organizing, and interpreting data

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Spatial and temporal statistics is a branch of applied statistics that emphasizes

- ❶ the context of the data,
- ❷ the spatial and time dependent relationship between data
- ❸ the different relative value and precision of the data.

Quantifying uncertainty

On the data sources side

- Confidence intervals
- Relevance
- Significance
- Correlation
- Causation
- Data Filters
- Biases identification